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  minimal degree $\mu(G)$: $\min\{\deg(x) : x \in G\}$

  e.g $\mu(\mathrm{Sym}(n)) = 2$, $\mu(\mathrm{Alt}(n)) = 3$ (for $n \geq 3$)

- $G_\omega := \{g \in G | \omega^g = \omega\}$
- block: $B \subseteq \Omega$ with either $B^g = B$ or $B^g \cap B = \varnothing$ for any $g \in G$

  block system: for finite Ω, $\Omega = B_1 \sqcup B_2 \sqcup \ldots \sqcup B_k$, $B_i^g = B_j$
- G is primitive $\iff$ G_ω is a maximal subgroup of G

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  ex:$\{1,2,3,4\} = \{1,2\} \sqcup \{3,4\}, G = \langle (1,3,2,4),(1,2) \rangle \cong D_8$

  block systems of G $\leftrightarrow$subgroups of G containing G_ω

  G is primitive on Ω if there is no non-trivial block system of G acting on Ω

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  Aschbacher 1989: Classification of primitive actions of classical groups

  O'Nan-Scott-Preager: Classification of finite primitive G (90s)

History for $\mu(G)$, G primitive

Babai 1981: If $\mathrm{Alt}(\Omega) \not\leq G,$ then $\mu(G) \geq 1/2\left(\sqrt{|\Omega|} - 1\right)$

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  2. $A_m^r \trianglelefteq G \leq S_m^r \rtimes S_r \leq \mathrm{Sym}(\Delta^r)$, $\Delta = \{k$-subsets of $[m]\}$ or $\Delta = [6]$

  3. $L^r \trianglelefteq G \trianglelefteq L_1^r \rtimes S_r \leq \mathrm{Sym}(\Delta^r)$, L is a simple group of Lie type over $\mathbb{F}_2$,

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Guralnick-Magaard 1998:

- if G is primitive with $\mu(G) < \frac{1}{2}|\Omega|$, then

  either 1. G is in LS.2 with $k \leq m/4$, $m=6$

  or 2. G is in LS.3 with $L_1 = O_m(2)$ with $m \geq 6$

Observations:

1. for $x \in G \leq \mathrm{Sym}(\Omega)$, $\mathrm{fix}_\Omega(x) \subseteq \mathrm{fix}_\Omega(x^m)$
2. for $x \in G$, and any block system $\mathcal{B}$

$$\Omega = B_1 \sqcup B_2 \sqcup \ldots \sqcup B_k$$

$$|\mathrm{fix}_\Omega(x)| / |\Omega| \leq |\mathrm{fix}_\mathcal{B}(x)| / |\mathcal{B}|$$

3. $\deg(x) / |\Omega| = 1 - |\mathrm{fix}_\Omega(x)| / |\Omega|$

fixed-point-ratio of x with respect to the action of G on Ω is

$$|\mathrm{fix}_\Omega(x)| / |\Omega|$$

Therefore,

$\min \{\deg(x) / |\Omega| | x \in G\} \Longleftrightarrow$ maximum fixed-point-ratio among

all prime-order elements of G respect to all primtive actions

$\mathbb{F}_q$: finite field, $q = p^f$ for prime p

$\mathrm{GL}_n(q)$: invertible $n \times n$ matrices over $\mathbb{F}_q$

$\mathrm{SL}_n(q)$: $n \times n$ matrices of determinant 1 over $\mathbb{F}_q$

$\mathrm{PSL}_n(q)$: $\mathrm{SL}_n(q) / Z(\mathrm{SL}_n(q))$

ex: $\mathrm{GL}_n(2) = \mathrm{SL}_n(2) = \mathrm{PSL}_n(2)$

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  $|x|$ is coprime to $p \iff x$ is diagonalizable over $\overline{\mathbb{F}_q}$;

  $|x|$ is a power of $p \iff$ all eigenvalues of x are 1

Elements of a prime order r, $r \neq p$

$x \in \mathrm{GL}_n(q), |x| = r \Longleftrightarrow$ eigenvalues of x are r-th root of unity, and $x \neq 1$

- Take a field $\mathbb{K} = \mathbb{F}_{q^i}$ s.t (TFAE)
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- $x \in \mathrm{GL}_n(q) \Longleftrightarrow x_{ij}^q = x_{ij}$

  the eigenvalues of x are invariant under $\mathrm{Gal}(\mathbb{F}_{q^i} / \mathbb{F}_q)$

  x is similar to $\mathrm{diag}(\Lambda_1, \ldots, \Lambda_k, I_e)$ over $\mathbb{F}_q$ with

  1. the characteristic polynomial of Λ_j is irreducible over $\mathbb{F}_q$,

  with eigenvalues $\{\omega, \omega^q, \ldots, \omega^{q^{i-1}}\}$, where $\omega \neq 1, \omega^r = 1$

  2. Λ_i is similar to the $\mathbb{F}_q$-linear map of $\mathbb{F}_{q^i}$ by multiplying by ω

$$\mathbb{F}_{q^i} \longrightarrow \mathbb{F}_{q^i} \colon \lambda \longmapsto \lambda w$$

Guralnick-Magaard induction

$H \leq \mathrm{Sym}(\Omega)$, $[H, H]$ is quasisimple

$\mu_H(g)$ maximum fixed-point-ratio of g among all faithful action of H

- key 1: $H = A \times B$, $g = (a, b) \in H$, $[B, B]$ quasi-simple with $b \notin Z([B, B])$
- key 2: $P = U \rtimes L$, $U \cong \mathbb{F}_p^k$ (U elementary abelian of order p)

  $[L, L]$ quasi-simple, $[L, L]$ acts irreducibly on U, $g \in P$ s.t $gU \notin Z(P/U)$

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  then for an action of P on Ω, either

  1. $|\mathrm{fix}_\Omega(g)|/|\Omega| \le \mu_{P/U}(gU)$
  2. for some $u \in U$, we have $C_L(u) \le P_\omega$

Example $G = L_n(2)$ by induction

Case 1: $\dim(C_V(g)) \geq 1$, so $g \in P_1$

$P_1 = U \rtimes L$ (the stabilizer of 1-dim subspace)

$U = \left(\begin{array}{c|c} 1 & * \\ \hline 0 & I_{n-1} \end{array}\right) \cong \mathbb{F}_2^{n-1}$, $L \cong \left\{ \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & A \end{array}\right) \middle| A \in L_{n-1}(2) \right\} \cong L_{n-1}(2)$

Case 1.1 If g is similar to an element in L over $\mathbb{F}_q$

(conjugate in $G = \mathrm{GL}_n(2)$ to an element in L)

then by induction $|\mathrm{fix}_\Omega(g)|/|\Omega| < 1/2$

Case 1.2. If g is not conjugate in G to any element in L

then $|g| = 2$,

g is similar to $\mathrm{diag}\left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \ldots, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right)$ (since $|g| = 2$)

so by key 2, either $1.$ $|\mathrm{fix}_\Omega(g)|/|\Omega| \leq \mu_{P_1/U}(gU)$

or G_ω contains $C_L(u)$ for some $u \in U$,

note that $P = U \rtimes L$, with $U \cong \mathbb{F}_2^{n-1}$ and L acts transitively on

the non-zero elements in U, so WMA $u = \left(\begin{array}{c|c|c} 1 & 1 & 0 \\ \hline 0 & 1 & 0 \\ \hline 0 & 0 & I_{n-2} \end{array}\right)$, and

$$C_L(u) = \left\{ \left(\begin{array}{c|c|c} 1 & 0 & 0 \\ \hline 0 & 1 & 0 \\ \hline 0 & \alpha' & A \end{array}\right) | \alpha' \in \mathbb{F}_2^{n-2}, A \in L_{n-2}(2) \right\} \cong \mathbb{F}_2^{n-2} \rtimes L_{n-2}(2)$$

Subgroups generate by transvections:

J. McLaughlin 1989: If $\mathbb{K} \neq \mathbb{F}_2$, $\dim(V) > 2$, G is a subgroup generated

by transvections, and 1 is the only unipotent subgroup nomarlized by G, then

$G = G_1 \times \ldots \times G_t$ correspond to $V = V_1 \oplus \ldots \oplus V_t$ with

1. V_i invariant under G

2. $G_i|_{V_j} = 1$ for $i \neq j$, 3. $G_i|_{V_i} = \mathrm{SL}(V_i)$ or $\mathrm{Sp}(V_i)$

P.J Cameron 1989: for $\mathbb{K}=\mathbb{F}_2$ two more cases

1. $G_i|V_i = O(V_i)$

2. $G_i|V_i \cong \mathrm{Sym}(\Omega)$ with $|\Omega|\geq 5$, V_i is the irreducible permutation submodule of G_i

Now if $G_\omega \neq P_{n-1}, P_1$ (the stabilizer of $(n-1)$-dimensional subspace)

Since G_ω contain $C_L(u)$, the unipotent subgroup normalized by $C_L(u)$ is

$$\left(\begin{array}{c|c} I_2 & 0 \\ \hline * & I_{n-2} \end{array}\right),$$

$G_\omega = P_{n-2}$, thus $G_\omega = P_1, P_{n-1}, P_2, P_{n-2}$

Now $g \sim \mathrm{diag}\left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \ldots, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\right)$ so

1. $\dim(C_V(g)) = n/2$, there are $2^{n/2}-1$ 1-dim spaces fixed by g

and there are $(2^{n/2}-1)(2^{n/2-1}-1)/3$ 2-dim s.t g acts trivially on them

2. as a permutation on $\mathbb{F}_2^n$, g fixes $2^{n/2}$ vectors and acts as $2^{n/2-1}(2^{n/2}-1)$

2-cycles on remaining vectors

{2-dim subspaces s.t g acts as a transvection}$\leftrightarrow${2-cycles of g}

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- Totally there are $(2^n-1)(2^{n-1}-1)/3$ 2-subspace of $\mathbb{F}_2^n$

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  so $|\mathrm{fix}_\Omega(g)|/|\Omega| \leq \frac{3\times 2^{n/2-1}(2^{n/2}-1)}{(2^n-1)(2^{n-1}-1)} + \frac{(2^{n/2}-1)(2^{n/2-1}-1)}{(2^n-1)(2^{n-1}-1)} < \frac{1}{2}$

Case 2.

- g acts irreducibly on V
- The centralizer of g

  $\mathbb{F}_{2^n} \cong \mathbb{F}_2^n$ as an $\mathbb{F}_2$-linear space, multiply by any element in $\mathbb{F}_{2^n}$ is a $\mathbb{F}_2$-linear map
- Permutation character: $\pi(g) = \mathrm{fix}_\Omega(g)$

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  so $g \sim [\Lambda]$, the eigenvalues of g over $\mathbb{F}_{2^n}$ are $\{\omega, \omega^2, \ldots, \omega^{2^{n-1}}\}$

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  $|\mathrm{fix}_\Omega(g)|/|\Omega| \leq \frac{1}{m+1} + \frac{|C_{L_n(2)}(g)|^{1/2}}{m}$,

  with m the minimum degree among all complex characters

Case 3: $g \sim [\Lambda_1, \ldots, \Lambda_{n/d}]$ with $\dim(\Lambda_i) = d$

for $n \leq 5$ by GAP with permutaiton character, one exceptional case

$G = L_4(2) \cong A_8$, with $G_\omega = A_7$ (the action is on 8 pts)

for $n \geq 5$, g is conjugate to an element in $L_d(2) \times L_{n-d}(2)$

$\{\left(\begin{array}{c|c} A & 0 \\ \hline 0 & B \end{array}\right) | A \in L_d(2), B \in L_{n-d}(2)\}$

By key 1 and induction either 1. $|\mathrm{fix}_\Omega(g)|/|\Omega| < \mu_{L_{n-d}}(g|_{L_{n-d}(2)})$

or 2. G_ω contains $L_{n-d}(2)$

In 1, either $|\mathrm{fix}_\Omega(g)|/|\Omega| < 1/2$, or $n = 6, d = 2,$ G_ω contains A_7

In 2, the unipotent subgroup normalized by $\langle g, L_{n-d}(2)\rangle$ is

$\left(\begin{array}{c|c} I_d & 0 \\ \hline * & I_{n-d} \end{array}\right)$, or $\left(\begin{array}{c|c} I_d & * \\ \hline 0 & I_{n-d} \end{array}\right)$, or 1

so $G_\omega = P_d, P_{n-d}$ (stabilizer of d or $n-d$-dim subspaces)

or $d = n - d$ and $G_\omega = L_d(2)^2 \rtimes S_2$

So $G_\omega = P_d, P_{n-d}$, $L_d(2)^2 \rtimes S_2$, or $n = 6$ and $G_\omega = \mathrm{Sp}(6,2)$

$g \sim [\Lambda_1, \ldots, \Lambda_{n/d}]$ with the characteristic polynomial of Λ_i irrducible

over $\mathbb{F}_2$, with eigenvalues in $\mathbb{F}_{2^d}$ $\{\omega, \omega^2, \ldots, \omega^{2^{d-1}}\}$, and $\mathbb{F}_2[\omega] = \mathbb{F}_{2^d}$

Λ_i similar to multiplying by ω on $\mathbb{F}_{2^d}$

- For any $d-$dim subspace W fixed by g, $g|_W = \Lambda_i$
- Similarly, when g is similar to a scalar matrix in $\mathrm{GL}_{n/d}(2^d)$, g fixes most $n-d$ dim subspaces

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- Similarly, when g is similar to a scalar matrix in $\mathrm{GL}_{n/d}(2^d)$, g

  fixes most $n-d$ dim subspaces

  when $n = 2d$ and g is similar to a scalar matrix in $\mathrm{GL}_2(2^d)$,

  then g fixes most direct decompostion of V to 2 d-dim subspaces

If $n=6, d=2$ $(G=L_6(2))$ and $G_\omega = \mathrm{Sp}(6,2)$

Then the action of G on Ω is euiqvalent to an orbit of G acting

on $\Lambda^2(V) = \mathrm{Span}_{\mathbb{F}_2}\{e_i \otimes e_j - e_j \otimes e_i | 1 \leq i, j \leq 6\}$

So $|\mathrm{fix}_\Omega(g)| = |C_{\Lambda^2(V)}(g) \setminus \{0\}|$

$g \sim [\omega, \omega^2, \omega, \omega^2, \omega, \omega^2]$ over $\mathbb{F}_4$, with $\omega^3 = 1$

Take $\{e_1, \ldots, e_6\}$ eigenvectors of g over $\mathbb{F}_4 \otimes_{\mathbb{F}_2} \mathbb{F}_2^6 = \mathbb{F}_4^6$ then

$C_{\mathbb{F}_4 \otimes_{\mathbb{F}_2} \Lambda^2(V)}(g)$

$=\mathrm{Span}_{\mathbb{F}_4}\{e_1 \Lambda e_2, e_1 \Lambda e_4, e_1 \Lambda e_{6,} e_3 \Lambda e_2, e_3 \Lambda e_4, e_3 \Lambda e_{6,} e_5 \Lambda e_2, e_5 \Lambda e_4, e_5 \Lambda e_6\}$

and $\dim(C_{\Lambda^2(V)}(g)) = \dim(C_{\mathbb{F}_4 \otimes_{\mathbb{F}_2} \Lambda^2(V)}(g)) = 9$

So $|\mathrm{fix}_\Omega(g)| / |\Omega| = \frac{2^9 - 1}{[G : G_w]}$

Burness Guranick 2022: 1.(G, x, Ω) with $\mathrm{fix}_\Omega(x) > \frac{1}{r+1}$ 2. lists of G with $\mu(G) < \frac{2}{3}|\Omega|$